



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Defect Detection System Decision using Computer Vision

ITA0512– Computer Vision

ASSIGNMENT REPORT

Submitted by HEMASRINITHA P

1. Problem Understanding and Formulation

What is the problem?

A manufacturing company wants to deploy an automated computer vision-based defect detection system on a high-speed production line. The system must identify subtle manufacturing defects with more than 95% accuracy. It must also adapt when product appearance, lighting conditions, orientation, and manufacturing batches vary. The central engineering decision is whether to use traditional computer vision methods or a deep learning approach.

Expected outcomes

The expected outcome is an end-to-end inspection solution that can acquire product images or video, preprocess and enhance the input, detect and classify defects, localize defective regions, evaluate performance, and produce a quality-inspection result. The selected approach should satisfy the accuracy target while remaining practical for industrial deployment.

Available information/data

The assignment provides two candidate solution families. Traditional computer vision can use filtering, thresholding, edge detection, contour analysis, morphological operations, shape/texture features, and template matching. Deep learning can use CNNs, transfer learning, YOLO, Faster R-CNN, or EfficientNet/MobileNet-based classifiers. Product images or video, together with labelled examples of normal and defective products, are the main data required for a trainable solution.

Assumptions

The production line provides images or video of products with sufficient visual information for inspection. Defects have visual characteristics that can be learned or represented from the available data. For the deep learning solution, enough representative training and validation examples are assumed to be available, including variations in lighting, orientation, product appearance, and manufacturing batches.

Constraints

The system must achieve more than 95% detection accuracy, operate reliably on a high-speed production line, adapt to product and environmental variations, and consider computational and training requirements. The solution should also be scalable and suitable for real-world deployment rather than only laboratory testing.

2. Application of Course Knowledge

Relevant computer vision principles

The solution applies image acquisition, image representation, preprocessing, enhancement, segmentation, feature/representation extraction, object detection, classification, localization, and performance evaluation. These stages convert raw production-line imagery into an actionable quality-inspection decision.

Preprocessing and enhancement

The proposed preprocessing stage includes resizing, noise removal, illumination correction, and filtering. Segmentation can be used when the product or suspected defect must be separated from the background. These operations help provide a more consistent input for subsequent detection.

Feature and representation analysis

For a traditional system, shape and texture features, edges, contours, and template information can represent defect characteristics. In a deep learning system, the model learns useful visual representations from training images, reducing the need to manually define every feature.

Algorithms and models

Traditional alternatives include thresholding, edge detection, contour analysis, morphological operations, and template matching. Deep learning alternatives listed in the assignment include CNN, transfer learning, YOLO, Faster R-CNN, and EfficientNet/MobileNet-based classifiers. The final design selects a deep learning approach because the problem requires both high accuracy and adaptability.

Engineering concepts

Important engineering concepts are accuracy, adaptability, computational cost, training requirements, robustness, scalability, and deployment suitability. The decision is therefore treated as a multi-criteria engineering evaluation rather than a choice based only on raw detection speed.

Reasoning for model selection

A practical design can use a transfer-learning-based detector, with YOLO as a strong candidate for high-speed visual inspection. The model can be trained and validated using representative product images and tested on unseen variations. If the task is classification-only,

an EfficientNet or MobileNet-based classifier can be used after the defect region has been isolated. The final choice should be confirmed by validation metrics and production constraints.

3. Solution / Design / Methodology

Proposed end-to-end pipeline

Product Image/Video Acquisition → Preprocessing → Image Enhancement → Defect Detection → Feature/Representation Extraction → Classification/Detection → Defect Localization → Performance Evaluation → Quality Inspection Report.

Stage 1 – Image acquisition

Capture product images or video frames from the high-speed production line. The acquisition setup should maintain a consistent view of the product while allowing the system to handle changes in orientation and appearance.

Stage 2 – Preprocessing

Resize images to the model input size, remove or reduce noise, correct illumination differences, and apply appropriate filtering. These steps aim to make the input more consistent across batches and lighting conditions.

Stage 3 – Defect detection and localization

Use a deep learning detector such as YOLO to identify defective regions and locate them with bounding boxes. The detection output can include the predicted defect class and confidence value. Localization is important because the inspection system should indicate where the defect occurs, not only whether a product is defective.

Stage 4 – Classification/decision

Classify the detected region or product into the required inspection categories, such as acceptable or defective, with defect type where labels are available. The final quality decision is generated from the model prediction and a selected confidence threshold.

Stage 5 – Performance evaluation

Evaluate the system on validation and test data. Accuracy should exceed 95% as required by the problem statement. Additional useful metrics include precision, recall, F1-score, detection confidence, inference time, and the number of correctly and incorrectly detected samples.

Alternative solution – Traditional computer vision

A controlled production environment could use thresholding, filtering, edge detection, contour analysis, morphological operations, and template matching. This approach is fast and computationally efficient, but it is less adaptive when product appearance, lighting, orientation, or manufacturing batches change.

Selected solution – Deep learning

Deep learning is selected as the preferred approach because the assignment requires greater than 95% detection accuracy and adaptation to significant product variations. Transfer learning

can reduce the amount of training needed compared with building a model entirely from scratch, while a detector such as YOLO is suitable when both speed and localization are important.

4. Use of Modern Tools

Tools and technologies

Python can be used for implementation, with computer vision and deep learning libraries for preprocessing, model training, inference, and visualization. TensorFlow or an equivalent deep learning framework can support model development. Git can be used for source-code versioning and project documentation. GPU or cloud computing resources may be used when required for model training.

Evidence of tool usage

The final implementation should provide source code, dataset information, training/testing outputs, performance metrics, sample detections, visualizations, and a README containing installation and execution instructions. These items also support the required GitHub submission.

5. Results and Validation

Validation plan

The dataset should be separated into training, validation, and testing portions. Training data is used to learn the model parameters; validation data is used to monitor model performance and tune the solution; and test data is reserved for final evaluation on unseen examples.

Performance measures

The main acceptance requirement is detection accuracy above 95%. Precision measures how many predicted defects are correct, recall measures how many actual defects are detected, and F1-score summarizes the balance between precision and recall. In an industrial setting, inference time and robustness to appearance and lighting changes should also be monitored.

Results reporting

A results table should report the selected model, accuracy, precision, recall, F1-score, inference time, and observations. Sample images should show normal products and detected defects with visual annotations. A confusion matrix or suitable chart can be included to demonstrate classification behaviour.

Important limitation

No experimental dataset, source code, or measured model output is supplied in the assignment brief. Therefore, this report does not invent numerical experimental results. The >95% requirement is treated as the acceptance target; actual measured values must be inserted after implementation and testing.

6. Analysis and Engineering Decision

Comparison of alternatives

Traditional computer vision is advantageous when the inspection environment is highly controlled because it can be fast and computationally efficient. However, its adaptability is limited when product appearance and environmental conditions vary. Deep learning requires sufficient training data, model training, and computational resources, but it is better suited to high accuracy and complex variations.

Trade-off analysis

The main trade-off is simplicity and computational efficiency versus adaptability and detection performance. Traditional methods reduce training requirements but can require carefully designed rules and features. Deep learning increases training and resource requirements but can learn complex representations and adapt better to changing visual patterns.

Decision

Deep learning is the recommended engineering decision for the proposed system. A transfer-learning-based detector, with YOLO as a practical candidate, should be evaluated first because the production line requires high accuracy, adaptability, and defect localization. A traditional

system may still be appropriate when the manufacturing environment is highly controlled and its measured performance satisfies the requirements.

Deployment considerations

Before production deployment, the model should be tested on unseen batches and different lighting and orientation conditions. The system should be monitored for performance changes, and the training dataset should be updated when new product variations or defect types appear. Computational requirements should be matched to the production-line inference speed.

7. Broader Considerations

Sustainability and environment

Automated inspection can support responsible production by identifying defective products earlier and reducing the movement of defective items through later production stages. This aligns with the assignment's SDG mapping to SDG 12 – Responsible Consumption and Production.

Industry and innovation

The proposed computer vision system applies AI-based automation to industrial quality inspection and supports SDG 9 – Industry, Innovation and Infrastructure.

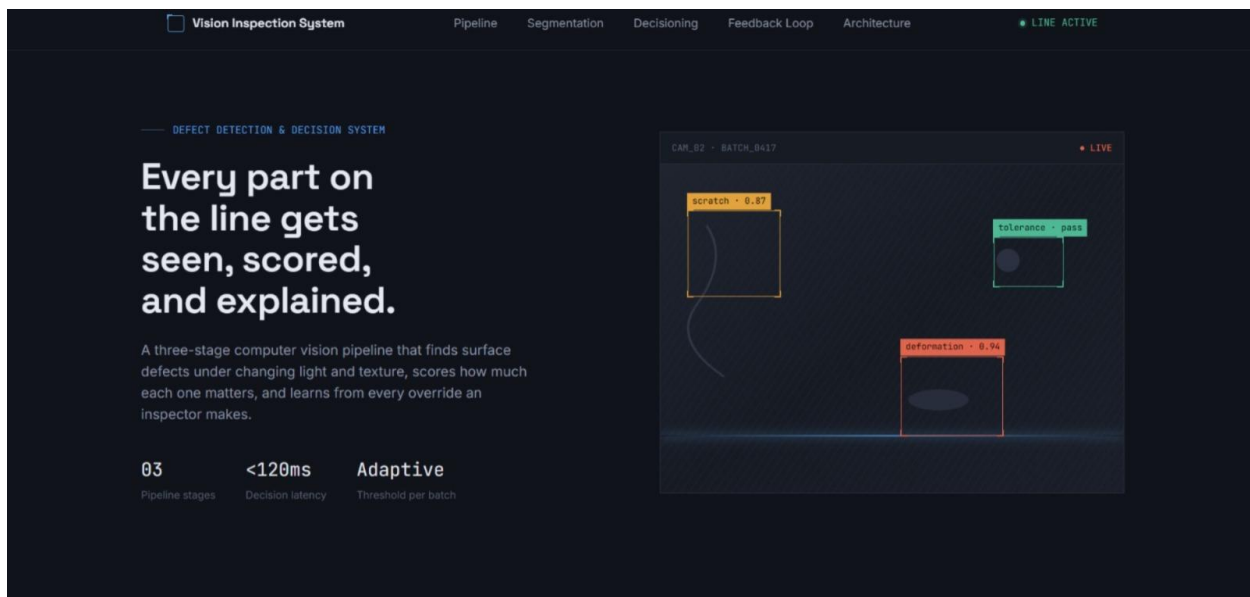
Economic and work considerations

Reliable automated inspection can improve production consistency and support efficient use of manufacturing resources. It also aligns with SDG 8 – Decent Work and Economic Growth by supporting modern industrial processes rather than relying only on manual inspection.

Safety, ethics, and professional responsibility

The system should be validated carefully before being used for quality decisions. False detections and missed defects should be monitored because inspection errors can have operational consequences. Model limitations and test conditions should be documented clearly, and deployment decisions should be based on measured evidence.

8 . IMPLEMENTATION



9. Conclusion

Summary

The assignment problem requires an automated computer vision system for high-speed industrial defect detection with more than 95% accuracy and adaptability to changes in product appearance, lighting, orientation, and manufacturing batches. Both traditional computer vision and deep learning approaches were considered.

Major finding

Traditional computer vision is fast and computationally efficient but is less adaptive to complex variations. Deep learning has higher training and computational requirements, but it is more appropriate when high accuracy and adaptability are essential.

Final solution

The recommended solution is a deep learning-based end-to-end inspection pipeline using preprocessing, defect detection, localization, classification, and performance evaluation. A transfer-learning-based detector such as YOLO is a suitable candidate for implementation, subject to validation against the >95% requirement.

Limitations and improvements

The current report does not contain measured experimental results because the assignment brief does not provide a dataset or implementation output. Future work should train and test the selected model on representative production data, compare it against a traditional baseline, optimize inference speed, and continuously evaluate performance on new manufacturing batches.

10. Student Reflection

Learning outcome

This assignment provides practical understanding of how computer vision concepts are converted into an industrial inspection pipeline. It highlights that selecting a model is not based only on accuracy; adaptability, computational resources, training requirements, robustness, scalability, and deployment constraints must also be considered.

What could be improved

With additional time and resources, the solution could be improved by collecting a larger and more diverse defect dataset, testing multiple deep learning models, tuning preprocessing and confidence thresholds, measuring inference performance on the intended hardware, and comparing the final deep learning system quantitatively with a traditional computer vision baseline.

11. References

Course assignment brief

Department of Computer Science and Engineering, ITA05 – Computer Vision, “Defect Detection System Decision using Computer Vision.”

Assignment format

Assignment Format document supplied for the assessment structure, including problem formulation, course knowledge, methodology, modern tools, validation, engineering decision, broader considerations, conclusion, reflection, and references.

Software/tool acknowledgement

Python and relevant computer vision/deep learning libraries are proposed as implementation tools. Any datasets, software libraries, pretrained models, cloud services, or AI-assisted tools actually used during implementation should be added to the final reference list and acknowledged in the submitted project documentation.

Approach Comparison Summary

Factor	Traditional CV	Deep Learning	Preferred for Problem	Reason		
Accuracy	Good in controlled settings	High potential	Deep Learning	Target is >95%		
Adaptability	Lower	Higher	Deep Learning	Handles complex variations better		
Computation	Lower	Higher	Depends on hardware	Production speed must be verified		
Training	Low/manual feature design	Requires training data	Deep Learning	Needed for changing appearance		
Robustness	Sensitive to conditions	Better with representative data	Deep Learning	Lighting/orientation/batch variation		
Localization	Rule-based possible	Direct detection/localization	Deep Learning	Useful for defect regions		

Deployment	Simple in controlled setup	More resource intensive	Deep Learning	Meets stated adaptability need		
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Git hub links

<https://github.com/HEMASRINITHA/ITA0512>